

Mélanie Debelgarric

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Education

- 2021–2024 **PhD in Ecophysiology**, *Université de Pau et des pays de l'Adour (UPPA)*, Saint-Pée-sur-Nivelle, France
- 2018–2021 **MSc in Ecological Systems Modelling (MSE)**, *Université Paul Sabatier Toulouse III*, Toulouse, France

Research Experience

- 2025 – 2027
(24 months) **Post-doctoral position**, *Université Claude Bernard Lyon 1, LBBE UMR 5558, Team IMPACT*, Lyon, France
▶ Development of a DEB-TKTD model for environmental risk assessment on the sentinel species *Gammarus fossarum*
Acquired skills: DEB-TKTD; Bayesian inference; Parameter estimation; Ecotoxicological data processing.
- 2021 – 2024
(36 months) **PhD Thesis**, *Université de Pau et des Pays de l'Adour, E2S UPPA, INRAE UMR ECOBIOP*, Saint-Pée-sur-Nivelle, France
▶ Unravelling the evolutionary potential of freshwater organisms facing climate change through Dynamic Energy Budget
Acquired skills: Dynamic Energy Budget theory; modelling in R & MatLab; life-history theory; "Pace-of-Life Syndromes (POLS)"; international seminars; scientific articles as first author ; PhD thesis dissertation in English.
- 2019 – 2020
(2 & 6 months) **Voluntary Internship & MSc Thesis**, *CNRS UMR LEFE & INRAE UMR RECOVER*, Toulouse & Aix-en-Provence, France
▶ Analyses of distribution in a Catfish population (*Silurus glanis*) showing seasonal aggregation behaviour
Acquired skills: Experimental setup (acoustic signals); data collection and processing; modelling in R (spatio-temporal trajectories); GIS; MSc thesis dissertation in English.
- 2019
(2 months) **MSc 1st year Internship**, *CNRS UMR CRCA*, Toulouse, France
▶ Modelling exploration and emergence of a pheromone network in the termite *Proconitermes araujo*
Acquired skills: Data processing; modelling in R (random walk, pheromone-induced movement); report writing.
- 2016
(2 & 2 months) **BSc voluntary Internships**, *CNRS UMR SETE*, Moulis, France
▶ Effect of temperature on the dispersion behaviour of ectotherms in the Pyrenees
Acquired skills: Data processing; modelling in R (basic statistical tests), field work including radio-tracking; organisation of an European meeting.

Schools & Trainings

- 2026
(5 days) **PREDITOX: School of Ecotoxicology & Predictive Modelling**, *Université Claude Bernard*, Lyon, France
[Link for more details](#)
- 2023
(10 days) **8th Thematic School on Dynamic Energy Budget Theory for Metabolic Organization**, *Louisiana State University*, Baton-Rouge, USA
[Link for more details](#)

Publications

- In prep. **Debelgarric, M.**, Geffard, O., Darde, J., Lopes, C. "Determination of heavy metals' modes of action on the reproduction investments in an amphipod using DEB-TKTD approach."
- In prep. **Debelgarric, M.** & Recapet, C. "Thermal Variation and Evolutionary Responses: Physiological Adaptations in *Daphnia magna* and *Salmo trutta*"
- 2026 **Debelgarric, M.**, Geffard, O., Felten, V., Lopes, C. "Molting else matters: an extended bioenergetic model for *Gammarus fossarum* that links molt and reproduction", *Ecological Modelling*, 522. DOI: 10.1016/j.ecolmodel.2026.111853
- 2025 **Debelgarric, M.** & Recapet, C. "Exploring physiological constraints on life-history traits using Dynamic Energy Budgets", *Ecological Modelling*, 501. DOI: 10.1016/j.ecolmodel.2024.110993
- 2017 Trochet, A., Le Chevalier, H., Calvez, O., Barthe, L., Isselin-Nondedeu, F., Picard, D., **Debelgarric, M.**, Pégourié, N., Rocher, R., Ribéron, A. "Postbreeding movements in marbled newts (Caudata, Salamandridae): a comparative radiotracking study in two habitat types". *Herpetologica*, 73(1), 1 - 9. DOI: 10.1655/Herpetologica-D-15-00072

Conferences & Oral Presentations

- Jul. 2026 Energy allocation and fitness: using DEB theory to model metal-induced trade-offs in a sentinel species, *Mathematical Modelling in Ecology and Evolution (MMEE)*, Cork, Ireland.
- Feb. 2026 Assessing the impact of heavy metals on *Gammarus fossarum* life-history traits using bioenergetics modelling, *Décryp'thèse 2026*, Lyon, France.
- Jan. 2026 Modelling the molting cycle in the sentinel species *Gammarus fossarum*: effects of heavy metals on reproduction, *Aquatic Ecotoxicology Workshop (GDREA2026)*, Banyuls-sur-mer, France.
- Mar. 2024 Unlocking the Drivers of Life-History Strategies in *Salmo trutta*: A Bioenergetic Modeling Approach, *International Workshop of PhDs and Post-doctoral Fellows on Anadromous Salmonids (NoWPaS2024)*, Cromarty, Scotland.
- Jun. 2023 An application of DEB theory to understand the co-variation of life-history traits in freshwater populations, *8th International Symposium on DEB Theory (DEB2023)*, Baton-Rouge, USA.
- May 2023 Modelling bioenergetics to understand the co-variation of life-history traits in freshwater populations, *28th European Meeting for PhD Students in Evolutionary Biology (EMPSEB28)*, Millport, Scotland.
- Mar. 2023 Modelling bioenergetics to understand the co-variation of life-history traits in salmonids, (*NoWPaS2023*), Oslo / Drøbak, Norway.
- Nov. 2022 Impact of global changes on freshwater populations: Use of the Dynamic Energy Budget model, *INRAE (JST LIFE)*, Thonon-les-Bains, France.
- Mar. 2022 Impact of global changes on salmonids population: use of the DEB model, (*NoWPaS2022*), Saint-Etienne-de-Baïgorry, France.
- Jun. 2019 Temperature dynamics in ant nests: observation, characterisation and theoretical framework, **Poster** *19th Insect Biology Colloquium (19CBI)*, Albi, France.

Grants

- 2023 **British Ecological Society (BES) Travel Grant**, to attend EMPSEB28, amount: 500 GBP
- 2023 **Fisheries Society of the British Isles (FSBI) Travel Grant**, to attend the DEB School and Symposium, amount: 1000 GBP
- 2023 **Université de Pau et des Pays de l'Adour (UPPA) Travel Grant**, to attend the DEB School and Symposium, amount: 800 EUR

Supervision & Responsibilities

- 2026 **Jury Member**, *Decryp'thèse*, Local competition in which PhD students at the University of Lyon explain their research topics.
- 2024 **Student Supervision**, *INRAE ECOBIOP*, Saint-Pée-sur-Nivelle, France
Supervised undergraduate internships students (1 MSc and 1 BSc) focused on allometric relationship determination using bioenergetic models.
- 2023-2024 **Organizing Committee**, *NoWPaS seminar*, Member of the committee with five other PhD students for two editions (in Norway and Scotland), in charge of international funding, activities, attendee selection and budget management.
- 2021-2024 **Teaching Assistant**, *Université de Pau et des Pays de l'Adour*, Anglet, France
Biostatistics for 1st year Biology students (*Total: 54h*).

Skills

- Programming R, MATLAB, Python, C++, QGIS, $\text{\LaTeX}$, Markdown, JAGS
- Modelling DEB Theory, Predictive models on ecotoxicology, Spatio-temporal and multivariate analyses, GIS.
- Statistics General Biostatistics, Frequentist and Bayesian inferences, ODEs.
- Fieldwork Telemetry (radio & acoustic tracking), electrofishing, sampling (amphibians, fish, insects).
- Lab work Animal caring and maintenance of experimental structures (amphibians, insects).
- Languages French (native), English (fluent), Italian (intermediate), German (beginner)

Interests & Activities

- Sciences Amateur astronomy (observation & optics) and naturalism.
- Sports Surfing, Rollerskating, Darts (sports licence), fitness.
- Music Bass guitar, Keyboards.
- Travel Numerous travel across 15 countries, including solo road-trips and backpacking.